

Jun-Kun Chen

Ph.D. Candidate in Computer Science | Controllable Generation | Visual World Models
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Research Focus

Ph.D. researcher in **diffusion-based controllable generation** and visual world models, with first/co-first-author papers at NeurIPS, CVPR, and 3DV and two Meta Reality Labs internships. Research spans image/video synthesis, long-horizon generation, multimodal conditioning, and **visual outcome prediction**; current work explores **bidirectional LLM-diffusion reasoning** and multimodal post-training.

Industry Experience

Meta Reality Labs

Research Scientist Intern (2025 continued part-time Aug.–Dec.)

May–Aug. 2023; May–Dec. 2025

Zurich, Switzerland / Bay Area, CA

- **2023**: Developed 3D-consistent diffusion editing with multi-view conditioning, structured noise, and self-supervised consistency training; published as *ConsistDreamer* at CVPR 2024.
- **2025**: Scaled training for a separate 3D/video-generation project to **128–256 GPUs** across multi-node clusters, enabling faster model and data iterations.

SpreeAI

AI Research Intern

May 2024 – May 2025

Remote, U.S.

- **Dress&Dance** (WACV 2027, under review): developed a multimodal conditioning mechanism and data-efficient multi-stage training for garment, identity, and motion control.
- **Virtual Fitting Room** (NeurIPS 2025): built anchored autoregressive generation for identity-consistent try-on videos of arbitrary duration, trained using only short clips.

Current Research: Generative World Models

Research vision: connect the complementary priors and capabilities of LLMs and diffusion models bidirectionally, then distill the joint loop into unified models.

Computer Use — Image-Diffusion World Model for UI Outcome Prediction

- Building a **SenseNova mixture-of-transformers (MoT) image-diffusion model** for instruction-conditioned UI outcome prediction and visual look-ahead in computer-use agents.
- Leading **diffusion post-training and joint Reasoner–Generator design**, including rollout-aware supervision and feedback from generated outcomes to agent reasoning.

Physics — Video-Diffusion World Model for Physical Events

- Post-training a **Cosmos3 video-diffusion world model** to predict physical events and outcomes from RGB history.
- Exploring a **teacher-guided Reasoner–Generator loop** that compares generated and target outcomes to adapt video reasoning and generation.

Robotics — Motion-Conditioned Video Diffusion World Model

- Built a synthetic prototype that predicts **robot–object interaction video** from a scene and prescribed robot motion.
- Designing **query-time proxy-to-target correspondence** and task-grounded evaluation of trajectories, contact, and outcomes across robots.

Selected Publications

1. Virtual Fitting Room: Generating Arbitrarily Long Videos of Virtual Try-On from a Single Image  **NeurIPS 2025**
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang [Project](#) | [Early preview \(arXiv, Sep. 2025\)](#)
2. Dress&Dance: Dress up and Dance as You Like It  **WACV 2027**, under review
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang [Project](#) | [Early preview \(arXiv, Aug. 2025\)](#)
3. HDEdit: Editing Videos and 3D Scenes with Video Diffusion Through Hierarchical Task Decomposition  **3DV 2026**
Yanming Zhang*, **Jun-Kun Chen***, Jipeng Lyu, Yu-Xiong Wang [Project](#) | [arXiv \(Mar. 2025\)](#)
4. ProEdit: Simple Progression is All You Need for High-Quality 3D Scene Editing  **NeurIPS 2024**
Jun-Kun Chen, Yu-Xiong Wang [Project](#) | [arXiv \(Nov. 2024\)](#)
5. Instruct 4D-to-4D: Editing 4D Scenes as Pseudo-3D Scenes Using 2D Diffusion  **CVPR 2024**
Linzhan Mou*, **Jun-Kun Chen***, Yu-Xiong Wang [Project](#) | [arXiv \(Jun. 2024\)](#)
6. ConsistDreamer: 3D-Consistent 2D Diffusion for High-Fidelity Scene Editing  **CVPR 2024**
Jun-Kun Chen, Samuel Rota Bulò, Norman Müller, Lorenzo Porzi, Peter Kontschieder, Yu-Xiong Wang [Project](#) | [arXiv \(Jun. 2024\)](#)

*Equal contribution. Preferred/publication name: **Jun-Kun Chen** (J.-K. Chen); legal name and earlier papers: Junkun Chen / J. Chen.

Education

University of Illinois Urbana-Champaign

Ph.D. Candidate in Computer Science; Advisor: Prof. **Yu-Xiong Wang**

Expected Dec. 2026

Urbana, IL

Tsinghua University

B.Eng. in Computer Science, Yao Class (IIS); Major GPA: 3.94/4.0

2017 – 2021

Beijing, China

Technical Skills

Models, Tools & Evaluation: PyTorch; Python; C++; controllable image/video diffusion; visual world models; multimodal conditioning and post-training; Transformers; vision-language models; data curation; benchmark design; quantitative and human evaluation.

Selected Research Contributions

Long-Horizon & Multimodal Video Generation — VFR and Dress&Dance

- Developed a **multimodal conditioning mechanism** for garment, text, layout, pose, identity, and motion controls; reduced FVD by 20–59% versus a Kling 1.6 try-on-and-animation pipeline across Internet and captured evaluations.
- Built anchored autoregressive generation for **minute-scale, temporally consistent video** from short training clips; led all three VBench consistency/smoothness metrics on 30–50 s hard-motion evaluations against FramePack and Kling 2.0.

General-Purpose Diffusion Editing — HDEdit and ProEdit

- Designed **hierarchical task decomposition** that separates instruction fulfillment from content preservation, combining LLM-based scheduling with interpolation-based refinement for training-free video and scene editing.
- Developed tuning-free preservation control and a **progressive editing workflow** with preview and edit-strength control, enabling high-fidelity geometry and appearance editing.

Consistent Scene and 4D Generation — ConsistDreamer and Instruct 4D-to-4D

- Created context-rich multi-view conditioning, **3D-consistent structured noise**, and self-supervised warping consistency for high-fidelity diffusion editing across views.
- Extended 2D diffusion to long-term dynamic scenes through **anchor-aware attention** and optical-flow-guided sliding windows.

Geometric Visual Representations — NeuralEditor and PointTree

- Built point-based radiance-field and relaxed k-d-tree representations for editable scene geometry and **transformation-robust point-cloud encoding**.

Additional Publications

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| 7. SceneCraft: Layout-Guided 3D Scene Generation  | NeurIPS 2024 |
| 8. Contrastive Learning Relies More on Spatial Inductive Bias Than Supervised Learning: An Empirical Study  | ICCV 2023 |
| 9. NeuralEditor: Editing Neural Radiance Fields via Manipulating Point Clouds  | CVPR 2023 |
| 10. PointTree: Transformation-Robust Point Cloud Encoder with Relaxed K-D Trees  | ECCV 2022 |
| 11. Is Self-Supervised Contrastive Learning More Robust Than Supervised Learning?  | ICML 2022 Workshop |
| 12. TorchDrug: A Powerful and Flexible Machine Learning Platform for Drug Discovery  | arXiv preprint, 2022 |
| 13. RNNLogic: Learning Logic Rules for Reasoning on Knowledge Graphs  | ICLR 2021 |
| 14. Pre-training Assisted Scene-aware Dialogue Generation  | DSTC8 Workshop at AAAI 2020 |

Earlier Experience

Mila – Quebec AI Institute

Research Intern; knowledge-graph reasoning with Prof. Jian Tang

Feb. – Aug. 2020

Montreal, Canada

Baidu, Natural Language Processing Department

Research / Development Intern; scene-aware dialogue generation (DSTC8 top-3 system)

July 2019 – Jan. 2020

Beijing, China

Competitions, Leadership & Coursework

Additional Competition Distinctions: CTSC 2017 (7th); CTSC 2016 Gold (8th); NOI Winter Camp 2016 Gold (3rd).

Leadership & Problem Setting: Problem setter for NOI, CTSC, ICPC, and CCPC events; Vice Chair, Tsinghua Student Association of Algorithms and Competitions (2018–2019).

Selected Coursework (Generative Models): Deep Generative Models (A+); Generative Models for Biomedicine and Life Sciences (A).

Vision & 3D: 3D Vision (A+); Deep Learning for Computer Vision (A); Computer Vision (A); Computational Photography (A+).

Learning: Meta-Learning (A+); Machine Learning (A).